

Program – 7: Deploy a Web Application to Kubernetes using Minikube

Description

This experiment demonstrates deploying an Nginx web server on a Kubernetes cluster using a Deployment and exposing it using a Service in Minikube.

Prerequisites

- Minikube installed
- kubectl installed

Step 1: Start Minikube

minikube start

Step 2: Check Minikube Status

minikube status

Step 3: Create Deployment (nginx-

deployment.yaml) apiVersion: apps/v1 kind:

Deployment metadata:

name: nginx-deployment spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:

metadata:

labels:

app: nginx

spec:

containers: -

name: nginx image:

nginx:latest ports:

- containerPort: 80

Step 4: Apply Deployment

kubectl apply -f nginx-deployment.yaml

Step 5:

Verify Deployment and Pods kubectl

get deployments kubectl get pods

Step 6: Expose Deployment as a Service kubectl expose deployment

nginx-deployment --type=NodePort --port=80 **Step 7: Access the Web**

Application minikube service nginx-deployment

This opens the Nginx web page in your

browser. **Step 8: Clean Up Resources** kubectl

delete pods --all kubectl delete svc --all kubectl

delete deployments --all

Expected Output

```
srikanths@Srikanths-MacBook-Air ~> cd Desktop/Home/Rv/devOps/
srikanths@Srikanths-MacBook-Air ~/D/H/R/devOps> ls
1RV24MC105_SRIKANTH.pdf  cap/                program7/            program9/
5thjan/                  program10/          program8/
srikanths@Srikanths-MacBook-Air ~/D/H/R/devOps> cd program7
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> ls
app.js      Dockerfile          package-lock.json  service.yaml
deployment.yaml  node_modules/      package.json
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl apply -f deployment.yaml
deployment.apps/kube-demo-deployment created
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kube-demo-deployment-b667756dc-4x4wx 0/1     ContainerCreating   0           7s
kube-demo-deployment-b667756dc-n5hbd 0/1     ErrImagePull        0           7s
nginx                                   1/1     Running             0           25m
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl get services
NAME      TYPE      CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes ClusterIP  10.96.0.1     <none>       443/TCP    94m
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl expose pod nginx --type=NodePort --port=8080
service/nginx exposed
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl get services
NAME      TYPE      CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes ClusterIP  10.96.0.1     <none>       443/TCP    95m
nginx     NodePort   10.110.74.235 <none>       8080:31495/TCP 10s
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kube-demo-deployment-b667756dc-4x4wx 0/1     ErrImagePull        0           46s
kube-demo-deployment-b667756dc-n5hbd 0/1     ErrImagePull        0           46s
nginx                                   1/1     Running             0           26m
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program7> █
```

Nginx default welcome page displayed in the browser.



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Result

Successfully deployed and accessed a web application on Kubernetes using Minikube.